

1M1B × IBM SKILLSBUILD

AI for Sustainability Virtual Internship

MealCycle AI

AI-Powered Food Waste Prevention & Surplus Decision Support System

Designed for IBM watsonx.ai & IBM Granite 3.0 (IBM BOB Stack)

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Primary SDG: SDG 2 , 12 ,13 — Responsible Consumption and Production

Aligned with: UN Sustainable Development Goal 12: Responsible Consumption and Production (Target 12.3)

Abstract

College canteens routinely over-prepare food because they lack a systematic way to combine attendance, day-of-week patterns, exams, holidays, events, and historical consumption trends into a single preparation estimate. MealCycle AI is a decision-support prototype built for the 1M1B AI for Sustainability internship that addresses this gap without overstating what a beginner-built AI system can responsibly claim.

The system combines a trained Linear Regression demand-prediction model (evaluated at MAE $\approx$ 25 portions, $R^2 \approx$ 0.86 on held-out synthetic data) with an explainable rule-based recommendation layer, a What-If scenario simulator, and three generative-AI workflows: prompt-engineered entity extraction from free-text staff notes, Retrieval-Augmented Generation for food-safety guideline lookup, and AI summarization of weekly trends. Throughout, the system is designed as decision support only — it never makes an autonomous decision, and it never determines whether food is safe to eat.

Introduction

Food waste in institutional settings is a well-documented but under-addressed contributor to Sustainable Development Goal 12 (Responsible Consumption and Production). Unlike household food waste, institutional over-preparation is a planning problem: the same kitchen, roughly the same menu rotation, and a demand signal that actually is predictable from a handful of known factors — if someone takes the time to model it.

MealCycle AI was developed to test whether a lightweight, explainable AI system — one buildable by a beginner BTech student without deep learning or cloud infrastructure — could meaningfully support that planning decision, while staying honest about its limitations.

Problem Statement

College canteens/messes prepare food based mostly on rough daily estimates or fixed quantities, without a systematic way to account for factors like day-of-week patterns, exams, holidays, events, and historical consumption trends. This mismatch between preparation and actual demand leads to avoidable over-preparation — of food, water, energy, and money — or, less often, under-preparation. Canteen staff currently lack a lightweight, explainable tool to combine these factors into a single, defensible preparation recommendation.

Institutional canteens in colleges and universities prepare hundreds of meals daily. Due to manual guesswork, canteen supervisors are estimated to over-prepare food by 15% to 25% on average, while under-preparation leads to sudden meal shortfalls during unexpected student attendance spikes. Food preparation currently relies on subjective intuition, and managers struggle to account for multi-variable factors such as upcoming placement drives, exam seasons, day-of-week attendance decay, and menu popularity. This results in:

- **Economic Loss:** Thousands of dollars spent on unconsumed ingredients.
- **Environmental Harm:** Organic food waste decomposing in landfills, releasing potent greenhouse gases (CH_4 and CO_2).

SDG Alignment

Primary SDG: SDG 12 — Responsible Consumption and Production, specifically Target 12.3, which calls for halving per-capita global food waste at the retail and consumer level by 2030. MealCycle AI targets the institutional food-service segment of this target — canteens are a consumption/production node where over-preparation directly causes avoidable waste.

Secondary alignment: SDG 2 (Zero Hunger), through more efficient use of prepared food resources, and SDG 13 (Climate Action), through reduced resource and energy waste associated with overproduction. These are mentioned as context; they are not the project's primary focus.

- **IBM AI technology integration:** The generative-AI workflows are designed around IBM Granite 3.0 8B Instruct via the IBM watsonx.ai runtime (IBM BOB platform), providing enterprise-grade NLP and RAG safety guardrails. The current prototype status is described under Limitations.
- **Carbon footprint metric:** Every 1 portion of wasted prepared food is estimated to generate approximately 0.8 kg of CO₂e emissions (including agricultural inputs, transport, and decomposition).

Existing Gap

Academic research on institutional food waste exists — for example, large multi-year datasets covering daily meal counts across school canteens, and studies distinguishing between food prepared-but-not-served versus served-but-not-eaten. However, these are research datasets tied to specific institutions and study designs; none are available as a ready-to-use public dataset matching an Indian college canteen's context (menu rotation, exam periods, campus events). No lightweight, freely available decision-support tool combining prediction, explainable recommendation, and scenario simulation for this specific context was identified.

Proposed Solution

MealCycle AI provides five integrated modules:

- Demand Prediction — a trained regression model estimates tomorrow's likely consumption from context (day, menu, attendance, exam/event/holiday flags).
- Surplus Analyzer — compares prepared vs. consumed quantities after service and categorizes surplus risk.
- Recommendation Engine — explainable, rule-based logic that adds a transparent safety buffer (+6%) on top of the model's prediction.
- What-If Simulator — lets staff compare a baseline plan against an adjusted scenario (e.g. lower attendance, smaller batch) before committing, with the financial (\$) and carbon footprint (CO₂e) impact modeled before cooking.
- Dashboard — sample-week metrics and an AI-generated plain-language weekly digest.

Three additional AI workflows were layered on top, chosen deliberately from the internship's suggested toolkit rather than included by default:

- Prompt engineering + entity extraction — a structured prompt converts a free-text staff note into the exact fields the model needs, always shown to the user for review before use.
- Retrieval-Augmented Generation (RAG) — a small set of guideline snippets is retrieved by keyword match and passed to an LLM, which answers food-safety-procedure questions only from that retrieved text.
- Summarization — the week’s raw numbers are turned into a short, plain-language digest for a canteen manager.

Methodology

Development approach

The project was built iteratively: dataset design and generation, model training and honest comparison across candidate algorithms, rule-based recommendation logic, a working browser-based prototype, and finally the three generative-AI workflows above — each added only after the previous layer was functioning and verified.

AI architecture

Data flows one direction, with a human decision at the end: historical/synthetic data → trained regression model → predicted demand → rule-based recommendation (prediction + safety buffer) → staff review and final decision. The prediction and recommendation steps are deliberately separate, explainable layers rather than a single opaque output.

Five-step human-centered pipeline

MealCycle AI follows a strict five-step human-centered pipeline, from a staff shift note or manual form through to a supervisor’s final approval:

Step	Stage	Output
Input	Staff Shift Note / Manual Form	Informal note or structured form entries
1	Entity Extractor (IBM Granite 3.0)	Validated structured model parameters
2	Multiple Linear Regression Model	Forecast base demand
3	Safety Buffer Rule Engine (+6%)	Calculates the recommended batch
4	Digital Kitchen Prep Docket	Barcode, timestamp & risk rating
5	Supervisor Final Approval	Human authorization to cook

The rule engine adds a fixed +6% safety buffer to the base forecast to produce the recommended batch. The result is issued as a digital kitchen prep docket carrying a barcode, a timestamp, and a risk rating, and every docket displays the explicit breakdown (base forecast + applied safety buffer). Nothing is prepared until the supervisor gives final approval.

Dataset

Prototype dataset created for demonstration purposes. No suitable public dataset matching this project’s exact feature set and context was found, so a synthetic dataset was generated: 180 days, with day-of-week, menu, and calendar effects (exam/event/holiday) built into the generation logic, plus realistic random noise, so a model trained on it has genuine patterns to learn rather than a trivial deterministic mapping.

Column	Description
date, day_of_week	Calendar context; day-of-week is a strong real-world demand driver
expected_students	Direct attendance estimate — the most direct demand signal
menu_type	Some menus are more popular, independent of headcount
event, exam, holiday	Binary flags for known demand-shifting circumstances
food_prepared, food_consumed	Used to compute surplus; food_consumed is the model's prediction target

food_waste was deliberately not stored as a separate synthetic column — it is calculated as prepared minus consumed, so the two numbers can never disagree with each other.

AI/ML Model

Three models were trained and honestly compared on an 80/20 train-test split (144 training days, 36 held-out test days): Linear Regression, a Decision Tree (max depth 4), and a Random Forest (200 trees, max depth 6).

Model	Test MAE / RMSE / R ²
Linear Regression (selected)	25.04 / 28.91 / 0.864
Decision Tree	29.25 / 39.51 / 0.745
Random Forest	27.45 / 34.77 / 0.803

Linear Regression was selected — it had both the lowest test error and the clearest explainability, since every prediction can be decomposed into named, interpretable coefficients (e.g. a holiday reduces expected consumption by roughly 185 portions relative to baseline). This directly supports the project’s Transparency requirement under Responsible AI.

Metric definitions (beginner-friendly)

MAE (Mean Absolute Error): on average, how many portions the prediction was off by. RMSE (Root Mean Squared Error): similar to MAE but penalizes large errors more. R² (R-squared): the proportion of variation in actual consumption the model explains, from 0 (no better than guessing the average) to 1 (perfect).

Regression model coefficients

The selected model is a Multiple Linear Regression (scikit-learn implementation). Its learned coefficients, which make every prediction traceable to named factors, are:

Model parameter / feature	Learned weight / coefficient	Operational impact
Base intercept (β_0)	+11.7532	Baseline constant
Student headcount (β_{students})	+0.8510 per student	Linear consumption scale
Campus event flag (β_{event})	+50.8643 portions	Demand spike during events
Exam season flag (β_{exam})	-60.6258 portions	Reduced turnout during exams
Holiday flag (β_{holiday})	-185.4583 portions	Significant attendance drop
Special / festive menu	+67.7284 portions	High turnout preference
Regular rotation menu	+39.1624 portions	Standard consumption

IBM AI Stack Integration (IBM watsonx.ai & Granite 3.0)

The extraction and retrieval workflows are designed around IBM Granite 3.0 8B Instruct via IBM watsonx.ai (IBM BOB platform), providing enterprise-grade NLP and RAG safety guardrails. The current prototype runs the same prompt design on a general-purpose LLM, as described under Limitations.

IBM Granite 3.0 kitchen note extractor

Supervisors often leave quick text notes rather than filling forms. The extractor is designed to use IBM Granite 3.0 8B Instruct via IBM watsonx.ai to parse informal text into validated JSON parameters:

Input text: "Tomorrow is Wednesday, expecting around 480 students with festive menu and a placement event."

Parsed output: {"expected_students": 480, "day_of_week": "Wednesday", "menu_type": "Special/Festive", "event": true}

IBM watsonx RAG policy assistant

To ensure surplus food handling complies with safety guidelines without AI hallucination, the guidelines are vector-indexed into discrete policy snippets (POL-01 to POL-05). Keywords from supervisor queries are matched against the indexed snippets using the IBM watsonx RAG architecture, and answers are strictly constructed from the retrieved policy snippets. The snippet text itself is placeholder example content (see Limitations).

Impact Assessment & What-If Simulator

The What-If Simulator dynamically models turnout variation (–40% to +40%) and lean-cooking adjustments (–30% to +30%), so staff can see the financial and environmental effect of a plan before cooking:

Impact	Formula
Financial impact	Cost Savings (\$) = (Baseline Prepared – Adjusted Prepared) × \$2.50 per portion
Environmental impact	Carbon Saved (kg CO ₂ e) = (Baseline Surplus – Adjusted Surplus) × 0.8 kg CO ₂ e per portion

The \$2.50 per-portion cost and the 0.8 kg CO₂e per-portion factor are assumed parameters used to compare scenarios; consistent with Expected Impact below, they are not measured savings.

Results

On 36 held-out test days (average actual consumption ≈ 366 portions), the selected model achieved a mean absolute error of approximately 25 portions — roughly a 7% typical error — and an R² of 0.86, meaning it explains about 86% of the day-to-day variation in consumption from the six input features. These figures are measured on synthetic data and should be treated as a demonstration of methodology, not a claim about real-world accuracy.

Responsible AI

Fairness

Predictions use averaged, aggregated consumption patterns — the model does not and cannot account for the fact that individual students eat different amounts. This is a known, documented limitation rather than a hidden assumption. Only aggregated headcount modeling is used: there is zero individual student tracking or dietary profiling.

Transparency

Every prediction is decomposable into the specific factors that produced it (day, menu, attendance, event/exam/holiday flags), because Linear Regression was chosen specifically for this property over less transparent alternatives. Every generated prep ticket also displays the explicit breakdown: base forecast plus the applied safety buffer.

Ethics

All AI outputs — predictions, recommendations, extracted entities, RAG answers, and summaries — are presented as decision support. None of them are auto-applied; a human always reviews and confirms before any number is acted on.

Privacy

No individually identifiable student data is used anywhere in the system — only aggregated daily headcounts and consumption totals. Zero Personally Identifiable Information (PII) is processed or stored.

Human Oversight

Canteen staff retain full responsibility for the final preparation decision. The system's role is strictly advisory: AI outputs are recommendations, and kitchen managers retain final authorization to cook.

Food Safety

The AI never independently declares food safe or unsafe to eat, reuse, or redistribute. The RAG-based guideline assistant is explicitly instructed to answer only from retrieved institutional text and to defer to authorized food-safety personnel whenever a question falls outside that text. Safety decisions follow the retrieved policy text.

Expected Impact

MealCycle AI aims to support more informed food-preparation decisions and may help reduce avoidable over-preparation over time, by giving canteen staff a consistent, explainable second opinion where previously there was only informal judgment. No specific quantity of food, water, cost, or emissions savings is claimed, since the prototype has not been deployed to measure a real before/after comparison — any such figure would be fabricated rather than measured.

Limitations

- The dataset is synthetic; real-world model accuracy is unverified and would need to be re-evaluated on actual canteen data before deployment.
- A small dataset (180 days) risks overfitting even with the train/test split used; more historical data would strengthen confidence in the results.
- The model cannot account for one-off events unless they are explicitly entered as a feature (e.g. an unannounced schedule change).
- Guideline text used by the RAG feature is placeholder example content, not a real institution's food-safety policy, and must be replaced before any real use.
- Entity extraction and summarization use a general-purpose LLM rather than IBM Granite, due to IBM Cloud account setup being outside this project's scope; the same prompt-engineering approach is designed to transfer directly to Granite via watsonx.ai.

Future Scope

- Replace synthetic data with a real canteen's logged attendance and consumption history.
- Swap the general-purpose LLM for IBM Granite via watsonx.ai, if the institution has IBM Cloud access.
- Add multimodal input — e.g. a photo of the serving counter at end-of-service — to help estimate leftover quantity, once labeled image data is available.
- Deploy the What-If Simulator and Dashboard as a lightweight Streamlit or hosted web app for daily use by canteen staff.

- **IBM Cloud deployment:** Scaling to a serverless deployment on IBM Cloud Code Engine.
- **Smart gate integration:** Integration with RFID campus turnstiles for real-time headcount updates.
- **Automated weather API:** Incorporating weather forecasts (e.g. rainy-weather headcount adjustments).
- **Multi-canteen management:** A centralized dashboard for multi-campus food management.

Conclusion

MealCycle AI demonstrates that a small, explainable machine learning model combined with clear rule-based logic, transparent AI-assisted workflows, and consistent human oversight can produce a genuinely useful daily decision-support tool — without overstating what a beginner-built prototype can guarantee. The project directly addresses SDG 12 by targeting institutional over-preparation, while treating Responsible AI not as a slide of principles but as constraints that shaped the system's actual architecture.

Combined with machine learning, IBM AI technology (IBM Granite 3.0 and IBM watsonx.ai) can support progress toward UN SDG 12.3 in educational institutions.